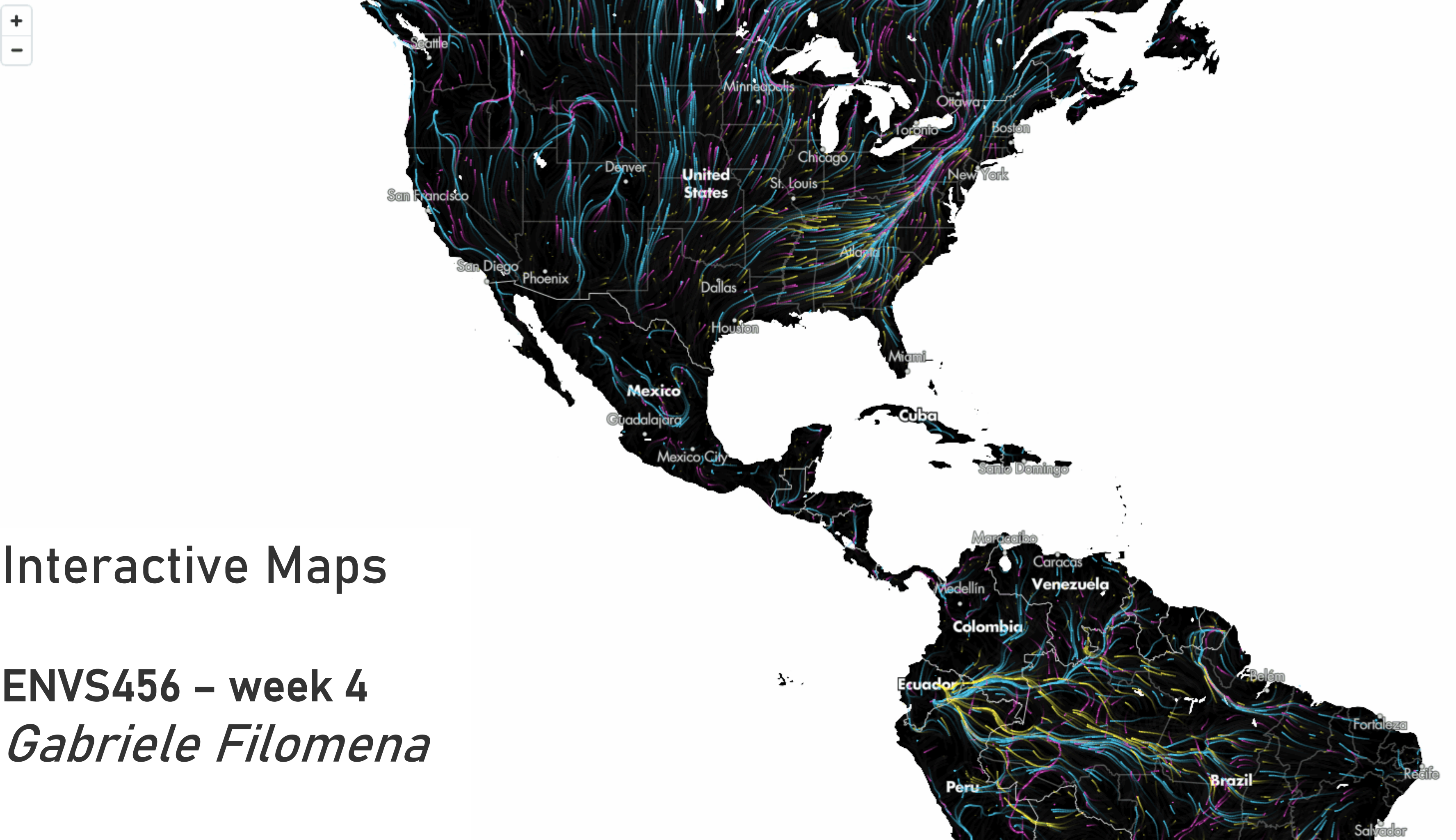




Interactive Maps

ENVS456 – week 4
Gabriele Filomena

Agenda

- Interactivity
- Interactive (Web) Maps
- Interactivity building blocks

Design Principles

- **Legibility:** Maps should be clear and straightforward
- **Accuracy:** maps should be a consistent representation of reality.
- **Aesthetic Appeal:** visually engage the audience

Important: Sourcing

Interactivity

Interactivity: *What*

- Ability to dynamically modify a visualisation
- Action/response as part of the experience

“[...] two-way flow of information, [...] responding immediately to the latter's [user's] input” ([Oxford English Dictionary](#))

Interactivity: *Why*

Munzner, 2014:

- Handle complexity.
- Cause the view to change.
- Support investigation at multiple levels of detail.
- Expand the capabilities of vis (many *idioms* depend on it).

Interactivity: *When*

- Too much to visualise all at once
- Both “big picture” and “detail” matter

“[when] seeing the dataset structure in detail is better than seeing only a brief summary of it”

Munzner (2014)

Interactive (Web) Maps

Interactive (Web) Maps

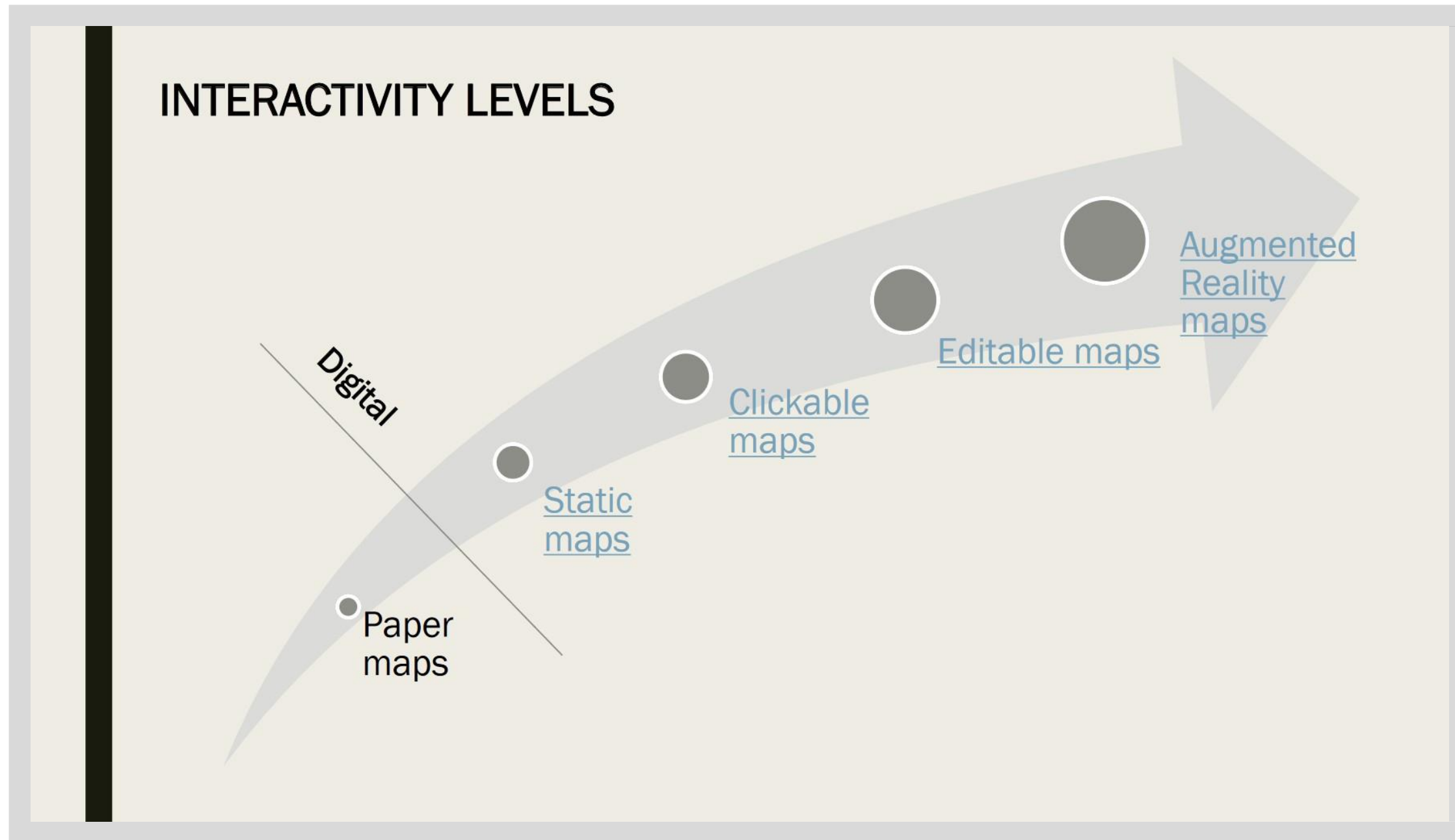


Image: A. Calafiore

Interactive (Web) Maps

- Efficient medium for high information throughput (Tufte)
- Maps as “windows” into large datasets

“A map of many maps”

Interactivity Building Blocks

Interactivity Building Blocks

- Filtering
 - Pan
 - Zoom
 - Subset
- Perspectives/Volumes
- Tooltips
- Split
- Animate

Filtering

- Widely used design choice in visualisation
- Reduction of the set of elements being displayed
- Discard geographically or attribute-based

Filtering: *Pan*

- **What:** Travels within a single scale
- **Use:** Segment a map geographically
- **Abuse:** Map is meant to focus on just a single region

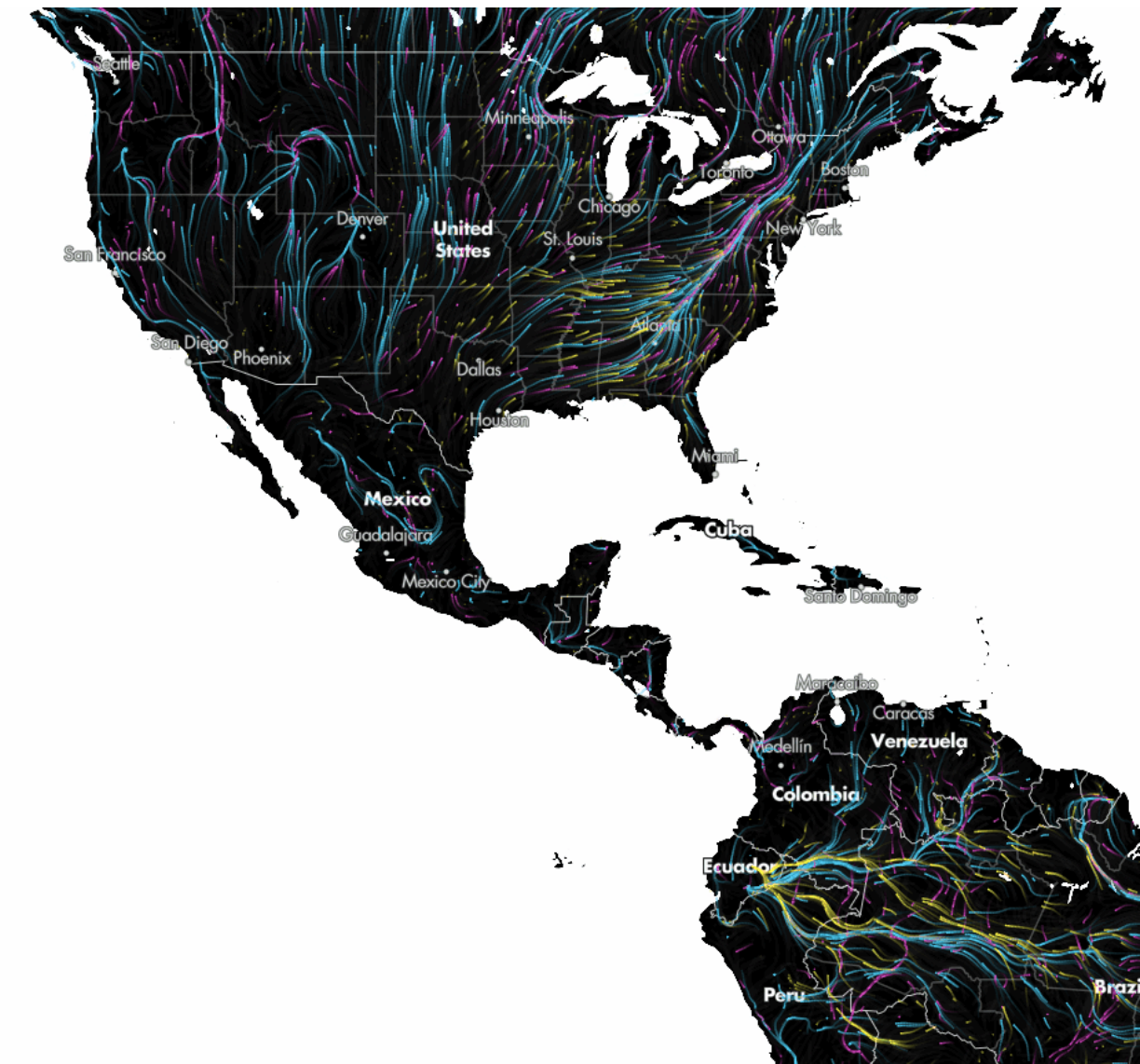


**MIGRATIONS
IN MOTION**

● Birds ● Mammals ● Amphibians

As climate change alters habitats and disrupts ecosystems, where will animals move to survive? And will human development prevent them from getting there?

This map shows the average direction mammals, birds, and amphibians need to move to track hospitable climates as they shift across the landscape. ↓



Filtering: *Zoom*

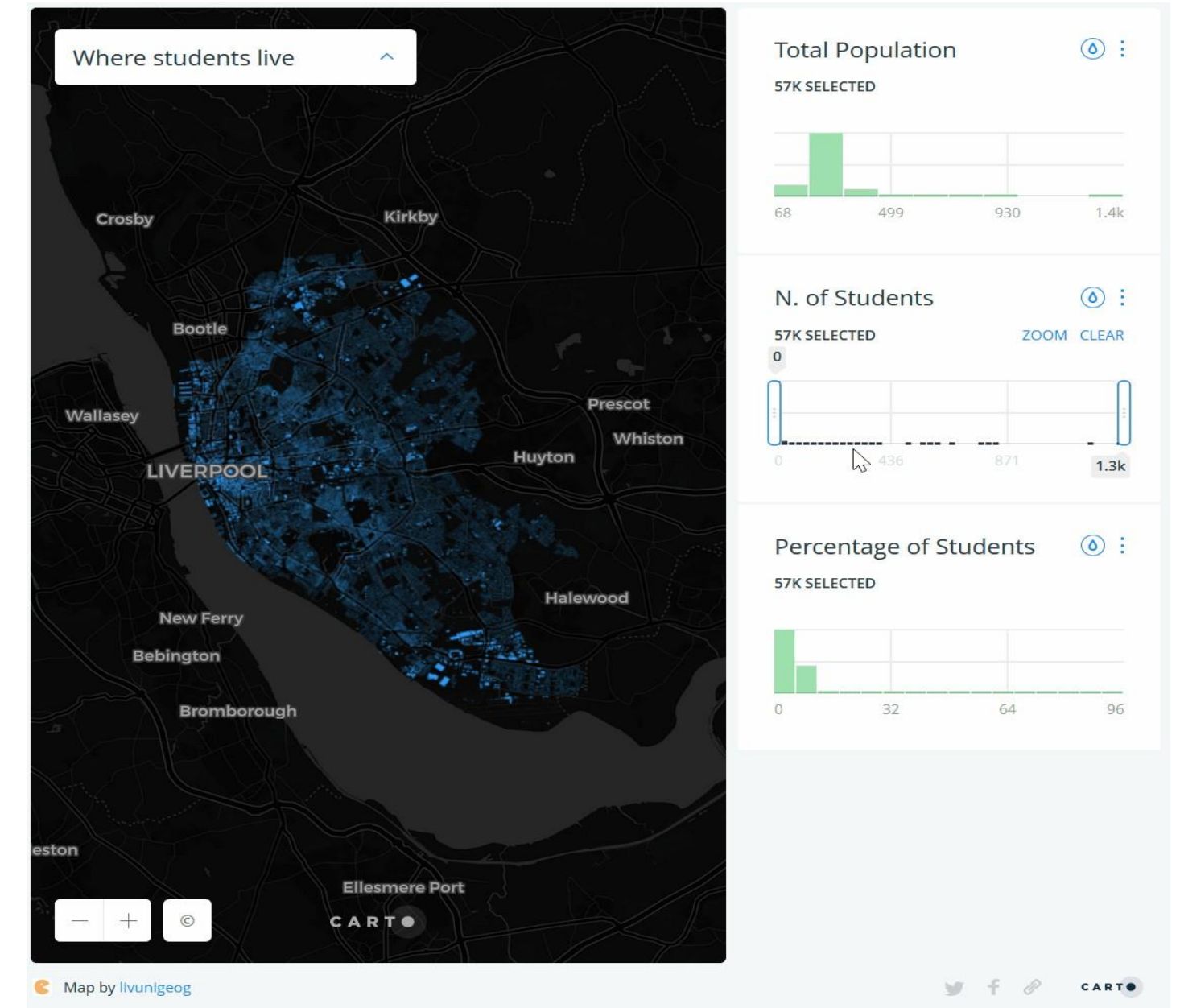
- **What:** “Travel” across scales
- **Use:** Present different degree of detail
- **Abuse:** Focus is on the global pattern



<https://medium.com/google-design/google-maps-cb0326d165f5>

Filtering: *Subset*

- **What:** Restrict data showed (by attribute)
- **Use:** Explore patterns by value or category
- **Abuse:** Focus is on the global pattern



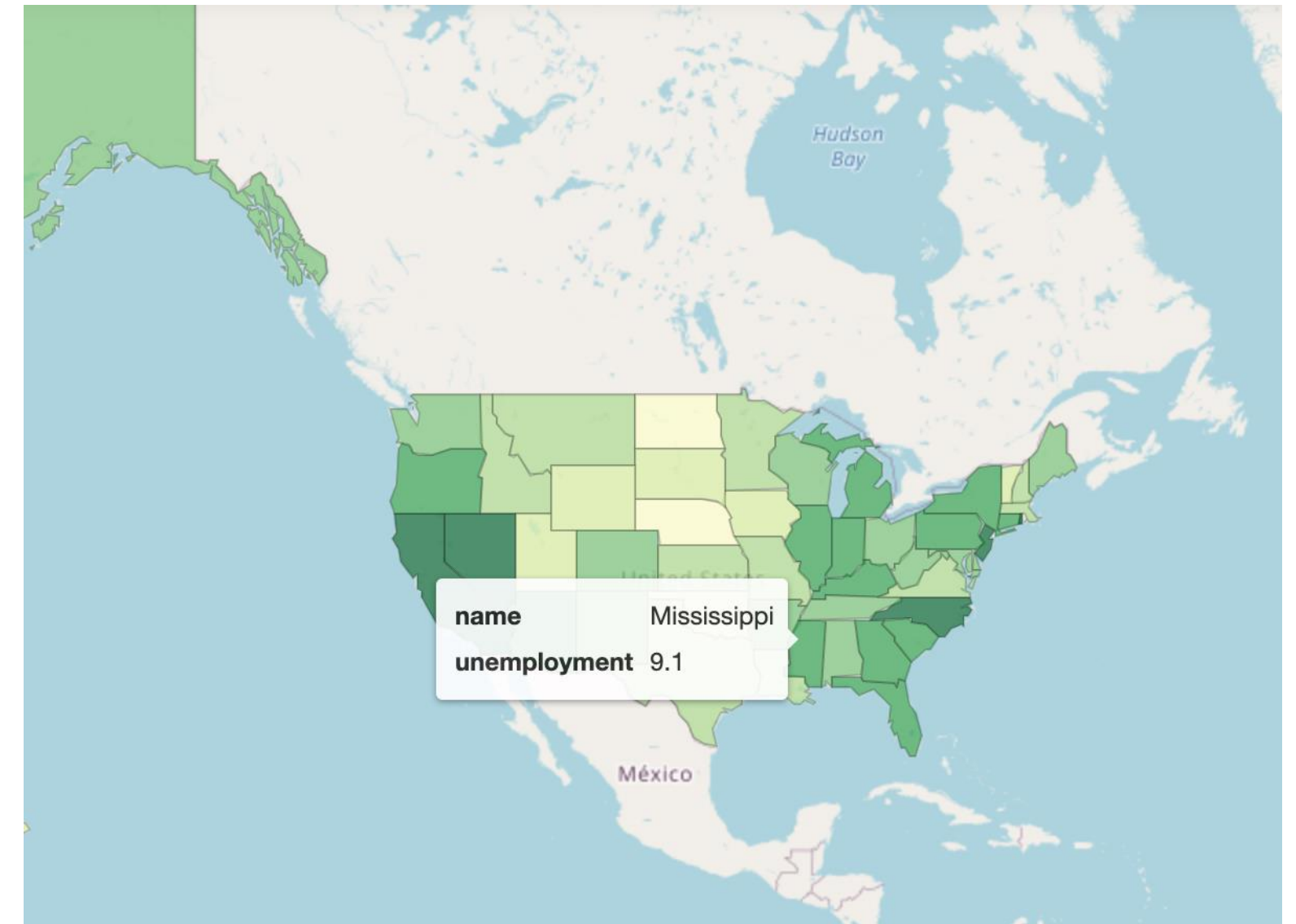
Volumes/Perspectives

- **What:** Add a 3rd dimension (or a 2.5d)
- **Use:** Volumes or perspectives are relevant
- **Abuse:** Any other case



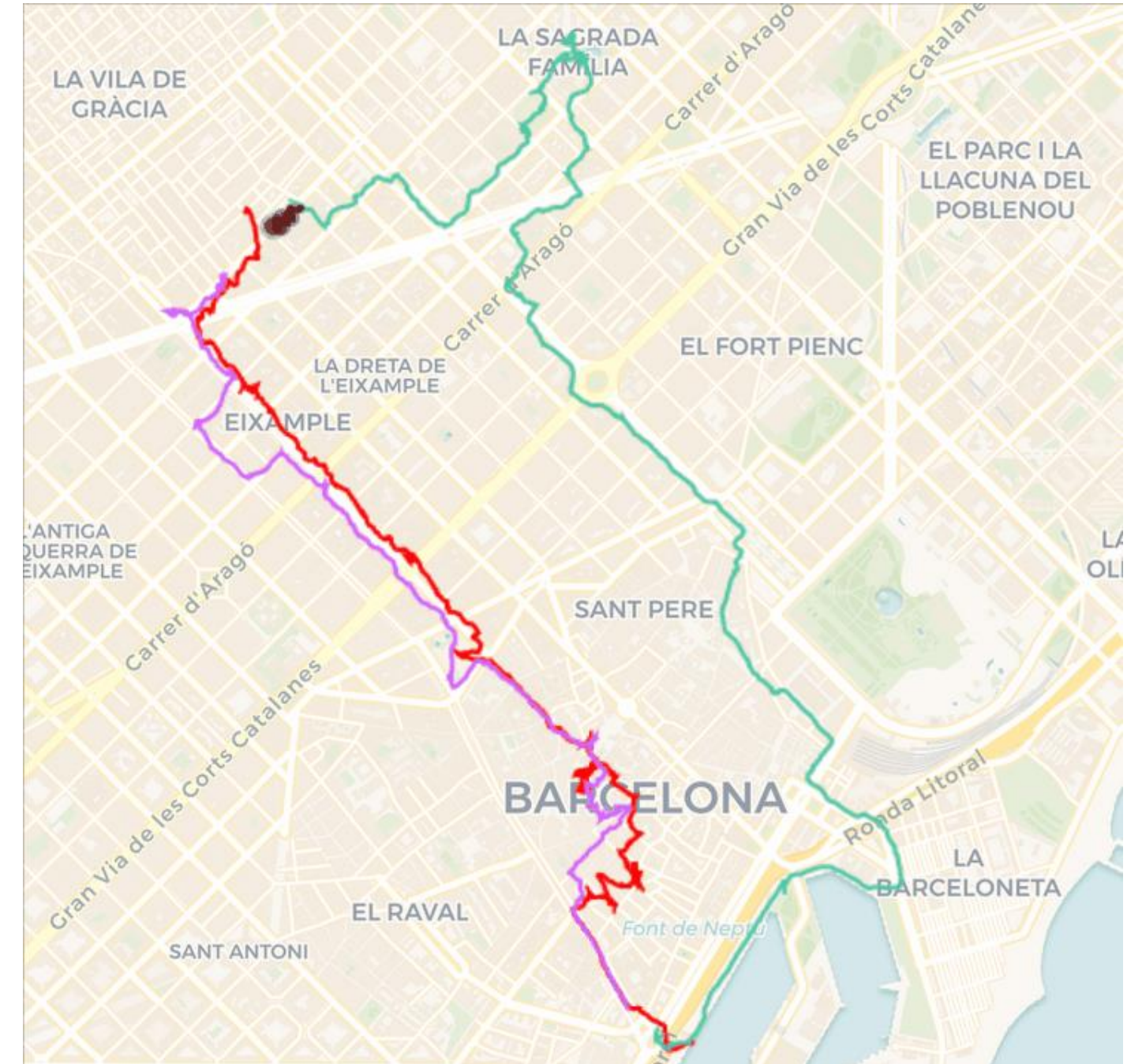
Tooltips

- **What:** Contextual (non-geo) information, on demand
- **Use:** Let the user explore the feature's attributes
- **Abuse:** Include too much data in the tooltip.



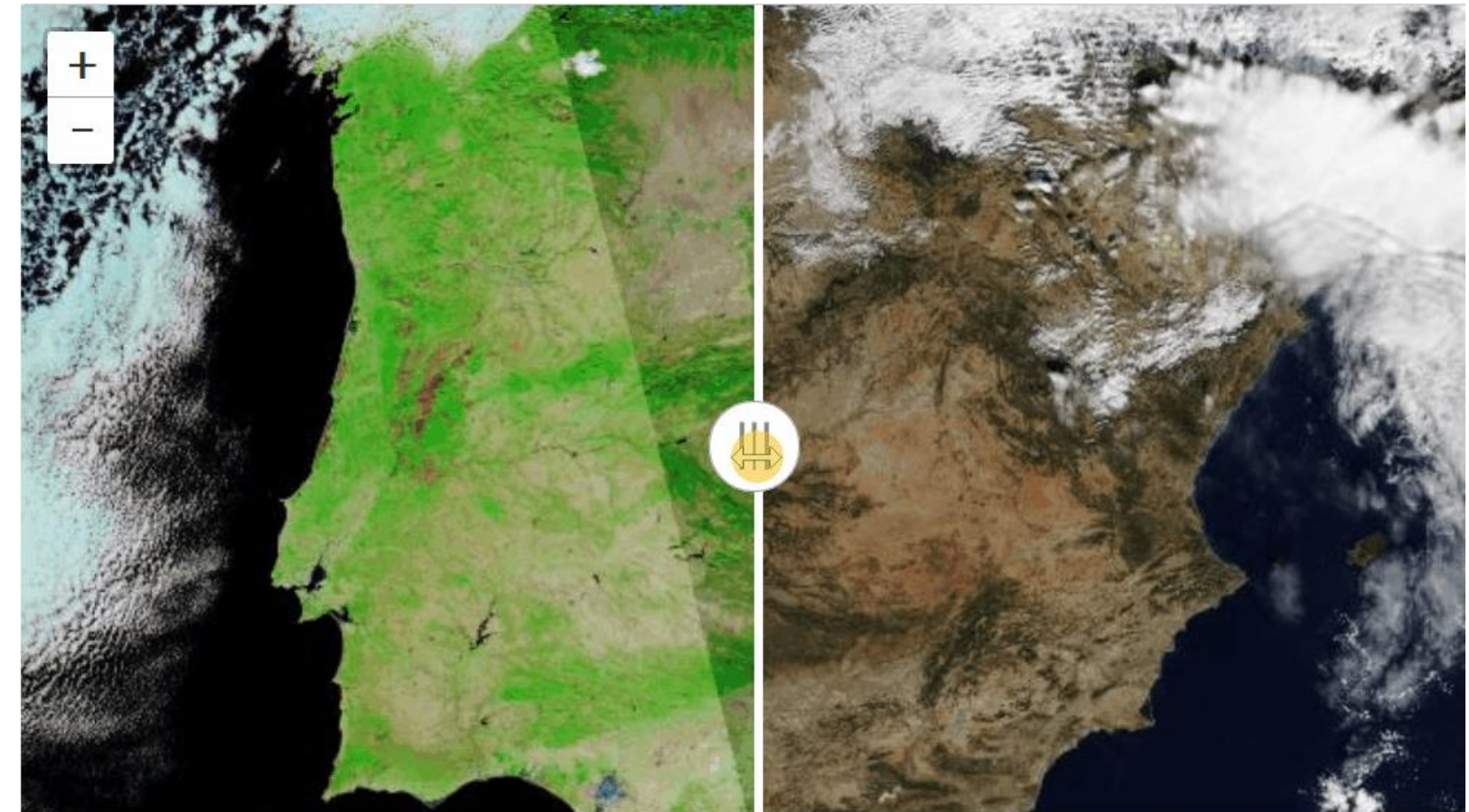
Animate

- **What:** Add temporal dimension through compilation of slices
- **Use:** Explore space-time Patterns
- **Abuse:** Communicate cross-sectional insights



Split

- **What:** Overlay two maps of the same location
- **Use:** Compare pattern changes
- **Abuse:** When you need to see the same location in both maps



Leaflet | Map data (c) [OpenStreetMap](#) contributors, Imagery provided by services from the Global Imagery Browse Services (GIBS), operated by the NASA/GSFC/Earth Science Data and Information System ([ESDIS](#)) with funding provided by NASA/HQ.

General tips

- Think about the experience first, then consider the technology.
- Avoid feature overload.
- Interactivity is not binary.

Let's talk about Assignment I

References

- Tamara, Munzner. *Visualization Analysis and Design*.
CRC Press, 2014